

CAT: preserve the intended spaces

Ask for the behavior, not just a neat result

01 / THE SAS EXAMPLE

```
data test;
  length first middle last $8
         name1 name2 $24;
  input first $ middle $ last $;
  name1 = cat(first, middle, last);
  name2 = catx(' ', first, middle, last);
  datalines;
John Q Public
Jane . Doe
Bob . Smith
;
run;

proc print data=test noobs;
run;
```

REPORTED TEST SETUP / 21 SEP 2026
Claude Sonnet 4.5 / AWS Bedrock GovCloud
Claude Code 2.1.278 / effortLevel=high / prompt v0.1.0

ATTACH SOURCE OR PASTE IT AFTER THE PROMPT. START FRESH EACH TIME.

WHAT YOU WILL PRACTICE

Distinguish preserving source behavior from silently cleaning the data.

CAT keeps blanks in these character fields. CATX trims them and separates nonblank values. Both behaviors are intentional. Inspect the spaces as well as the readable names.

EXPECTED RESULT

Row	CAT: name1	CATX: name2
1	John____Q_____Public__	John Q Public
2	Jane_____Doe_____	Jane Doe
3	Bob_____Smith__	Bob Smith

Each underscore represents one stored blank in name1. name2 hides trailing storage padding. Input periods become blank middle fields. Both stored result columns have width 24.

REPORTED FULL-REQUIREMENT MATCHES
Prompt A 0/3 | Prompt B 3/3

Reference values checked in Python; SAS execution pending.

Prompt A: leave choices open

High degrees of freedom / Notice the decisions the model makes for you.

COPY THIS PROMPT + THE SAS EXAMPLE

Convert this SAS code to Python.

Show code in this response. Do not create or run files, or read existing Python files.

REPORTED TRIAL RESULTS / 21 SEP 2026

Readable names can hide missing spaces

Run	CATX approach	Full match
A1	Join with spaces	No
A2	Two alternatives shown	No
A3	Filter periods + join	No

All three used unpadded + concatenation for CAT.

A3 matched readable CATX names; stored widths were missing.

A2 offered both CATX recipes seen in A1 and A3.

SOURCES: results-20260921.csv; trial-log-20260921.md; analysis-20260921.md

WHY CHOICES REMAIN OPEN

- 1

CAT keeps the stored blanks

All A runs reportedly joined raw inputs. The + operator is valid Python, but it cannot restore omitted padding.
- 2

A readable name is only one check

A3 filtered the input periods and got the readable names. Both stored results still needed width 24.
- 3

Alternatives need a decision

A2 offered two CATX recipes. Compare each with the required behavior before accepting either.

ANTI-PATTERN: ACCEPTING A NEAT NAME

Inspect visible blanks and stored widths. Ask for cleanup separately after checking the equivalent conversion.

Different Python methods can preserve the same behavior.

Prompt B: make the choices explicit

Low degrees of freedom / State the behavior, then ask for a check.

COPY THIS PROMPT + THE SAME SAS EXAMPLE

```
Convert this SAS CAT/CATX code to Python using pandas.
- Use first, middle, last in the original row order.
  Map the input periods to blank middle fields, not literal dots.
- Right-pad every input field to 8 characters with spaces.
- name1: concatenate the padded fields without trimming (CAT).
- name2: strip ordinary leading/trailing spaces from each field,
  skip blank fields, and join with one space (CATX).
- Store both result columns padded to width 24.
- Include assertions for both widths and all results below.
  For name1 checks, _ denotes one space:
  John__Q____Public__
  Jane____Doe____
  Bob_____Smith__
  name2, with trailing padding hidden:
  John Q Public; Jane Doe; Bob Smith.
- Print name1 with spaces visible and name2 in readable form.
  Keep stored values unchanged; explain the difference briefly.

Show code in this response. Do not create or run files,
or read existing Python files.
```

SAS REFERENCE

Reports describe six fresh parallel agents; complete transcripts and execution logs are absent.
Passing assertions are reported. Local reference checked; SAS run pending. Three runs do not guarantee future results.

WHY THESE DETAILS HELP

- 1 Separate CAT from CATX**
CAT preserves padded fields. CATX strips ordinary spaces, skips blanks and joins with one space.
- 2 Define the stored values**
Map input periods to blanks. Require input widths of 8 and result widths of 24; preserve row order.
- 3 Request a checkable result**
Name exact values and widths to assert. B1-B3 report padded CAT and strip/filter/join CATX.

REPORTED RESULTS / THREE B RUNS

Run	Full match	Assertions
B1	Yes	Passed
B2	Yes	Passed
B3	Yes	Passed

CHECK IT YOURSELF
Check exact CAT spaces and widths of 8 / 24.
Blank middle fields must add no CATX separator.